

GOBIKRISHNA P

AI / ML ENGINEER

Chennai, Tamil Nadu, 600130 | +91-76397 98140 | gobikrishnap.official@gmail.com | linkedin.com/in/gobikrishna-p/
github.com/Gobi200

SUMMARY

AI/ML Engineer with 1+ year of experience building and deploying Machine Learning and Generative AI applications using Python. Hands-on experience with Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP, prompt engineering, embeddings, and vector databases, alongside supervised and unsupervised machine learning model development. Skilled in data preprocessing, model training and evaluation, REST API integration, and deploying AI/ML solutions on AWS with Docker. Proven track record of collaborating with cross-functional teams to translate business requirements into scalable, production-ready AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

AI / Machine Learning: Machine Learning, Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Supervised Learning, Unsupervised Learning, Model Training & Evaluation

Generative AI / LLMOps: Prompt Engineering, Embeddings, Vector Databases, Semantic Search, RAG Pipelines, Document Intelligence

Python & Data Science: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Statistics, Data Preprocessing, Feature Engineering

Data Visualization / BI: Power BI, Tableau, DAX

Backend & APIs: RESTful APIs, FastAPI, Flask, API Integration, JSON

Databases: MySQL, PostgreSQL

Cloud & MLOps: AWS (ECR, ECS, ALB, API Gateway), Docker, CI/CD, Model Deployment, Git/GitHub

EXPERIENCE & CLIENT ENGAGEMENTS

Avasoft Pvt Ltd

Aug 2025 – Present

AI/ML Engineer

Chennai, India

- Design and develop Python-based AI/ML solutions for business applications, covering requirement analysis, model development, integration, and testing.
- Build Generative AI and LLM-based applications, including Retrieval-Augmented Generation (RAG) pipelines for document intelligence and knowledge retrieval.
- Develop data-processing workflows for document parsing, text chunking, embeddings generation, semantic search, and vector retrieval to support LLM applications.
- Apply prompt engineering techniques to optimize LLM outputs for accuracy, relevance, and business use cases.
- Prepare, clean, and transform structured data using Pandas, NumPy, and SQL to support machine learning model development.
- Train and evaluate machine learning models using Scikit-learn, applying appropriate evaluation metrics to validate model performance.
- Integrate AI/ML services with application backends through REST APIs, collaborating with cross-functional engineering teams.
- Follow Git/GitHub-based version control and collaborative development practices across the AI/ML development lifecycle.

Client Engagement: MLOps Pipeline Implementation

- MLOps Architecture & Tenant Isolation:** Designed and built an MLOps control plane to automate vendor onboarding, model lifecycle orchestration, and isolated cloud hosting of proprietary ML models.
- API Security & Access Control:** Implemented tenant-specific access controls using AWS API Gateway and custom middleware to generate, validate, and manage user IDs and expiring API keys, enforcing vendor-level data isolation.
- Administrative APIs & Service Discovery:** Built REST APIs for vendor-profile provisioning, containerized model deployment, and real-time service-discovery metadata.
- High Availability & Deployment Automation:** Automated AWS ECS deployments across multiple Availability Zones; configured ALB listener rules and weighted target groups for blue-green deployments and canary releases (staged traffic splits, e.g., 70/20/10/0) enabling zero-downtime updates.
- Service Health & Key Lifecycle Management:** Built background services to monitor ECS container states and automate API-key rotation and expiration.

Client Engagement: Data Analytics & Contamination Report

- **ETL Pipeline Development:** Built an end-to-end ETL pipeline using Power Query and SQL to extract, clean, and load sensor and inspection data for reporting.
- **Data Cleansing & Classification:** Processed inspection logs to identify and categorize physical contaminants, including metals, stones, paper, and plastics.
- **Data Modeling & Dashboard Development:** Designed a relational data model and interactive Power BI dashboards using DAX-based KPI calculations for contamination tracking.
- **Batch-Level Traceability:** Added dynamic filters and drill-down analysis enabling quality-assurance teams to investigate contamination trends by manufacturing batch.

Client Engagement: Logistics Freight Cost Prediction Engine

- **Legacy Process Automation:** Developed a machine learning regression model to replace a manual lookup-table process for delivery-cost estimation.
- **Data Preparation and Feature Engineering:** Built Python preprocessing workflows to clean historical logistics records, handle missing values, and encode categorical variables for model training.
- **Regression Model Deployment:** Trained and evaluated Scikit-learn regression models to predict shipping costs using distance, weight, and carrier-related features.
- **Model Validation and Optimization:** Performed hyperparameter tuning and model validation using MSE, RMSE, and R², achieving an R² score of 0.77 and an RMSE of 0.0412 on the test dataset.

Vihara Tech

Feb 2024 – Mar 2024

Data Analyst Intern

Hyderabad, India

- Analyzed 50,000+ sales records across 8 product categories using Python and SQL to identify trends and actionable business insights.
- Performed data processing and exploratory data analysis (EDA) to support data-driven business decisions.
- Translated analytical findings into insights that informed business growth and performance-improvement strategies.

Bolt IoT Inventrom Pvt Ltd

Jun 2023 – Jul 2023

Web Application Development Intern

Bengaluru, India

- Built a responsive image gallery web application using HTML, CSS, and JavaScript, demonstrating front-end development skills.
- Implemented interactive UI components and client-side functionality to improve usability and user experience.
- Applied web development fundamentals to translate design requirements into a functional application.

EDUCATION

Knowledge Institute of Technology

2021 – 2025

Bachelor of Technology (B.Tech) – Computer Science and Business System (CSBS) | CGPA: 8.02

Salem, India

Notre Dame of Holy Cross School, CBSE

2020 – 2021

Higher Secondary School Certification (HSSC) – Mathematics & Computer | Score: 83%

Salem, India

CERTIFICATIONS

- Elite Certification in Programming in Java - NPTEL
- SQL and Database Bootcamp Course - Udemy
- Data Analytics Course - IBM
- Professional Communication & Presentation Skills Course – TCS ION

HONORS & ACHIEVEMENTS

- Secured 1st Place out of competing colleges in the Sona College Technical Coding Contest (2023).
- Awarded 2nd Place (Runner-Up) in the Algorithmic Problem-Solving Contest at Sona College (2024).
- Earned Elite Grade Certification in Programming in Java from NPTEL (IIT proctored assessment).